Isospectrality defect of the nonlinear Schrödinger equation on looped networks: vertex curvature, dressed monodromy, and local integrable junctions

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Abstract

The focusing nonlinear Schrödinger equation supplies a Zakharov–Shabat transport on each edge of a metric graph. We derive an exact balance law for the trace of a travel-adapted loop transport: its time derivative is a sum of trace pairings with jumps of the time-Lax matrix at the visited vertices. For Kirchhoff coupling these jumps depend on derivatives along edges outside the loop. They can produce order-one spectral drift, but a nonzero jump alone does not imply drift: an explicit standing-wave family has nonzero vertex defects and conserved loop spectra. History-dependent vertex insertions restore simultaneous conjugation of dressed transports based at the same germ, including conservation of their Fricke commutator trace. This compensation does not by itself establish standard integrability. We derive local polynomial compatibility conditions, recover the Corrigan–Zambon defect with field-dependent diagonal coefficients, and obtain a shared-germ compatibility test and an exact recurrence at arbitrary spectral degree. These constrain local junction constructions without constituting a universal degree-three no-go theorem. Ring and theta simulations check the defect law and distinguish common-germ from different-germ dressing. A higher-gauge interpretation identifies the vertex defect with the lowest coherence operation; higher closure remains an additional problem.

1. INTRODUCTION

Solitons of the focusing nonlinear Schrödinger (NLS) equation carry signals through optical, photonic and matter-wave networks. On the line, inverse scattering maps NLS to spectral data with simple evolution [1]; on the circle, monodromy and finite-gap theory

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play this role [2, 3]. Caudrelier extended inverse scattering to star graphs and posed the finite-graph problem [4]. Variational and dynamical approaches also describe solitons on graphs [5–9].

We transport the Zakharov–Shabat auxiliary vector along a closed edge walk. This *travel-adapted loop transport*, also called loop holonomy here, uses the field in the direction of travel at the same spectral parameter λ. Reverse-edge transfer is not forward transfer inverted (Lemma 1); we assert no fundamental-group representation. For $\mathrm{SL}(2,\mathbb{C})$, *isospectrality* means constant eigenvalue multiset, equivalently constant trace. The *vertex defect* or *vertex curvature* F_v is the difference of arriving and departing time-Lax matrices; a *germ* specifies an incident edge and its travel direction.

Theorem 2 expresses the loop-trace derivative as a sum of transport–vertex-defect pairings. At Kirchhoff junctions, off-loop boundary derivatives determine the defect; these derivatives are not particle currents. Zero defects suffice for isospectrality, but nonzero contributions can cancel. Proposition 5 gives a family with increasing curvature and fixed spectra; soliton routing instead shows order-one drift.

Insertions $D_v = W_{\mathrm{out}} W_{\mathrm{in}}^{-1}$, where W_g is the germ's time-Lax evolution, give simultaneous conjugation for dressed loops sharing a departure germ (Theorem 7). Joint invariants, including the Fricke commutator trace, are conserved. This history-dependent compensation alone supplies neither local commuting charges nor an inverse-scattering solution.

We then test transmissions depending on instantaneous boundary fields and finitely many derivatives, polynomially in λ. This is *local Lax compatibility*; standard integrability requires additional Hamiltonian or inverse-scattering structure [10, 13]. We recover the Corrigan–Zambon jump defect with field-dependent diagonal coefficients, derive necessary shared-germ compatibility for three germs (Theorem 12), and give an arbitrary-degree recurrence (Proposition 13). These do not classify all nonlinear junctions, vertex degrees of freedom or higher coherence structures, or prove a no-go theorem by equation counting.

Ring and theta simulations test the identities and base-germ dependence (Sec. 6). A Sobirov-inspired basis control [9] distinguishes prescribed auxiliary gluing from identity gluing. Section 7 interprets vertex curvature as the lowest higher-gauge coherence operation; higher closure and local categorical integrability remain additional requirements.

2. SETUP

A. NLS on a metric graph and the Lax pair on an edge

A metric graph $\Gamma = (\mathcal{V}, \mathcal{E})$ has edges $e \cong [0, \ell_e]$ carrying fields $q_e(x,t)$ that satisfy the focusing NLS equation

$$\mathrm{i}q_t + q_{xx} + 2|q|^2 q = 0 \tag{1}$$

in the interior of every edge, and vertex conditions at $v \in \mathcal{V}$. Our numerical examples use the Kirchhoff condition (continuity of q and vanishing sum of the outward derivatives $\partial_e q$ over the edges incident to v), the standard self-adjoint coupling of quantum graphs [14]; Theorems 2 and 7 do not depend on the vertex condition, only on smoothness of q_e up to the ends of each edge.

Equation (1) is the compatibility condition of the Zakharov–Shabat pair [1]

$$\psi_x = U\psi, \qquad \psi_t = V\psi, \qquad U = \begin{pmatrix} -\mathrm{i}\lambda & q \\ -\bar{q} & \mathrm{i}\lambda \end{pmatrix}, \qquad V = \begin{pmatrix} -2\mathrm{i}\lambda^2 + \mathrm{i}|q|^2 & 2\lambda q + \mathrm{i}q_x \\ -2\lambda\bar{q} + \mathrm{i}\bar{q}_x & 2\mathrm{i}\lambda^2 - \mathrm{i}|q|^2 \end{pmatrix}, \tag{2}$$

in the sense that $U_t - V_x + [U,V] = 0$ is equivalent to (1) and its complex conjugate. Both matrices are traceless, so transfer matrices lie in $\mathrm{SL}(2,\mathbb{C})$.

B. Transfer along an edge; the orientation of the Zakharov–Shabat connection

For an edge traversed in the direction of increasing x the transfer matrix is the ordered exponential $T_e^{\rightarrow}(\lambda,t) = \overrightarrow{\exp}\int_0^{\ell_e} U(q_e(x,t),\lambda)\,dx$, so that $\psi(\ell_e) = T_e^{\rightarrow}\psi(0)$. Because the zero-curvature condition holds on the edge,

$$\partial_t T_e^{\rightarrow} = V_e(\ell_e)\,T_e^{\rightarrow} - T_e^{\rightarrow}\,V_e(0), \tag{3}$$

with $V_e(x) = V(q_e(x,t), \partial_x q_e(x,t), \lambda)$ evaluated at the two ends in the coordinate of the edge.

Lemma 1 (Orientation)**.** *Under the reflection $x \mapsto x' = \ell_e - x$ the connection one-form transforms as $U(q,\lambda)\,dx = U(-q(\ell_e - x'), -\lambda)\,dx'$. Consequently the inverse $(T_e^{\rightarrow})^{-1}$ is the transfer of the reflected field at spectral parameter $-\lambda$ and sign-reversed field, not the transfer of the reflected field at λ. The latter, $T_e^{\leftarrow}(\lambda,t) = \overrightarrow{\exp}\int_0^{\ell_e} U(q_e(\ell_e - x',t),\lambda)\,dx'$, satisfies (3) with the roles of the two ends exchanged and with $\partial_x q_e$ replaced by $-\partial_x q_e$ in V.*

Proof. $dx = -dx'$ and $-U(q,\lambda) = U(-q,-\lambda)$ entrywise. Since $q \mapsto q(\ell_e - x')$ solves (1) (the equation is reflection invariant), $T_e^{\leftarrow}$ is an ordinary edge transfer for the reflected field, whose derivative in its own coordinate is $-\partial_x q_e$. □

The lemma matters for graphs. Around a directed loop a shared edge may have to be traversed against its coordinate. Our same-parameter, travel-adapted convention uses $T_e^{\leftarrow}$, the transfer in the direction of travel; this is the convention under which a smooth ring is isospectral (Sec. 6, Fig. 2), whereas replacing the reversed transfer by $(T_e^{\rightarrow})^{-1}$ gives a different product. An ordinary connection does use an inverse upon path reversal; that construction is distinct from the travel-adapted spectral problem considered here.

C. Germs, loop holonomy, and the two time-Lax matrices at a vertex

A *germ* at v is an incident edge together with a direction of travel through v: arriving (the edge coordinate, in the direction of travel, points into v) or departing (it points away). For an arriving germ a we write $V_a^{\mathrm{in}}(v) = V(q(v), d_a^{\mathrm{in}}, \lambda)$ with d_a^{in} the derivative of the field in the direction of travel at v; for a departing germ b, $V_b^{\mathrm{out}}(v) = V(q(v), d_b^{\mathrm{out}}, \lambda)$. In terms of outward derivatives $\partial_e q$ (pointing away from v), $d_a^{\mathrm{in}} = -\partial_a q$ and $d_b^{\mathrm{out}} = \partial_b q$.

A directed loop $\gamma = (e_1, \dots, e_n)$ based at v_0 visits vertices $v_1, \dots, v_n = v_0$, departing v_{k-1} along e_k and arriving at v_k along e_k. With T_k the transfer of e_k in the direction of travel, the *loop holonomy with continuity gluing* is

$$H_\gamma = T_n \cdots T_1, \qquad H_\gamma^{(v_k)} := Q_k P_k, \quad Q_k = T_k \cdots T_1, \quad P_k = T_n \cdots T_{k+1}, \tag{4}$$

where $H_\gamma^{(v_k)}$ is the same loop based at v_k and $H_\gamma^{(v_0)} = H_\gamma$. The auxiliary vector is carried across each vertex unchanged; this is the gluing under which the loop holonomy is a function of the edge fields alone. The vertex curvature of the loop at v_k is

$$F_{v_k} := V_{e_k}^{\mathrm{in}}(v_k) - V_{e_{k+1}}^{\mathrm{out}}(v_k) \qquad (k = 0, \dots, n-1;\ e_0 \equiv e_n), \tag{5}$$

the jump of the time-Lax matrix between the arriving and the departing germ. Figure 1 fixes these conventions on the theta graph used throughout.

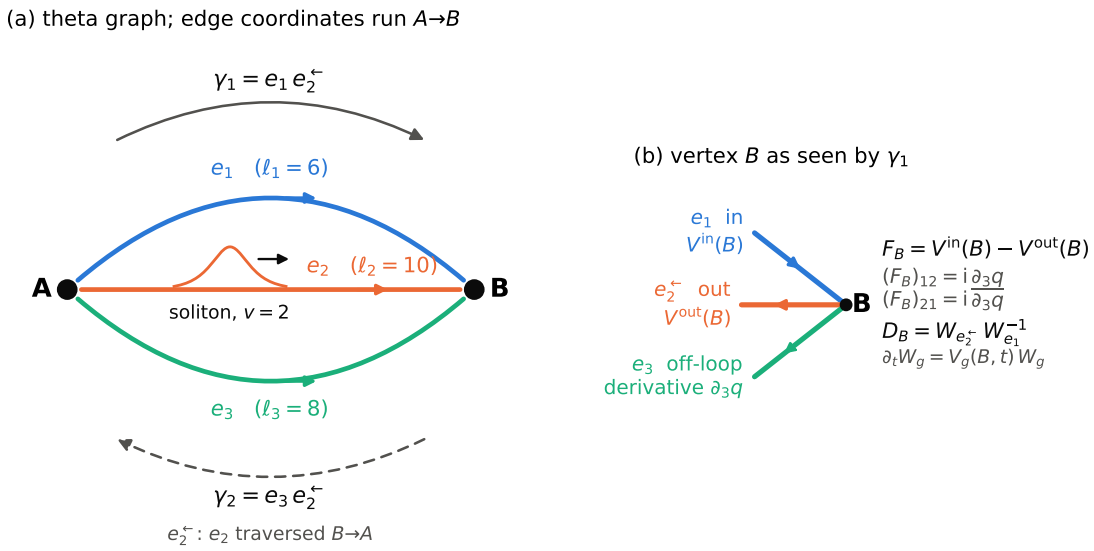


FIG. 1. Conventions. (a) Theta graph: edges e_1, e_2, e_3 of lengths $6, 10, 8$ have coordinates from A to B. Loop γ_1 leaves along e_1 and returns along $e_2^{\leftarrow}$, using the reflected field at the same λ, not inverse transport (Lemma 1); γ_2 returns along e_3. The soliton on e_2 moves toward B. For $\hat{\kappa}$, both loops are rewritten to depart A along e_2 (Remark 8). (b) At B, γ_1 arrives along e_1 with time-Lax matrix V^{in} and departs along $e_2^{\leftarrow}$ with V^{out}. Their difference F_B is formed from the outward derivative on off-loop edge e_3 by the Kirchhoff sum (Corollary 3). Dressing inserts $D_B = W_{e_2^{\leftarrow}} W_{e_1}^{-1}$, with W_g the germ's time-Lax evolution.

3. THE ISOSPECTRALITY DEFECT IS THE VERTEX CURVATURE

Theorem 2 (Defect law)**.** *Let the fields satisfy* (1) *on every edge of a directed loop* γ*. Vertex sums count crossings with multiplicity and use the arriving and departing germs at each crossing. Then*

$$\partial_t H_\gamma = [V^{\mathrm{out}}_{e_1}(v_0), H_\gamma] + F_{v_0} H_\gamma + \sum_{k=1}^{n-1} P_k F_{v_k} Q_k, \qquad \frac{d}{dt}\operatorname{tr} H_\gamma = \sum_{k=0}^{n-1} \operatorname{tr}\big(H_\gamma^{(v_k)} F_{v_k}\big). \tag{6}$$

In particular, $F_v = 0$ *at every crossing suffices for isospectrality. Along a given solution the trace is conserved precisely when the sum on the right vanishes; individual summands need not vanish.*

Proof. By (3) and Lemma 1, $\partial_t T_k = V^{\mathrm{in}}_{e_k}(v_k) T_k - T_k V^{\mathrm{out}}_{e_k}(v_{k-1})$ for every edge traversed in its

direction of travel. Hence

$$\partial_t H_\gamma = \sum_{k=1}^{n} P_k\big(V_{e_k}^{\mathrm{in}}(v_k)T_k - T_k V_{e_k}^{\mathrm{out}}(v_{k-1})\big)Q_{k-1} = \sum_{k=1}^{n} P_k V_{e_k}^{\mathrm{in}}(v_k)Q_k - \sum_{j=0}^{n-1} P_j V_{e_{j+1}}^{\mathrm{out}}(v_j)Q_j,$$

using $P_kT_k = P_{k-1}$ and $T_kQ_{k-1} = Q_k$. The terms with $1 \le k = j \le n-1$ combine into $P_kF_{v_k}Q_k$; the remaining ones are $P_nV_{e_n}^{\mathrm{in}}(v_n)Q_n = V_{e_n}^{\mathrm{in}}(v_0)H_\gamma$ and $-P_0V_{e_1}^{\mathrm{out}}(v_0)Q_0 = -H_\gamma V_{e_1}^{\mathrm{out}}(v_0)$, whose sum is $[V_{e_1}^{\mathrm{out}}(v_0), H_\gamma] + F_{v_0}H_\gamma$. Taking the trace kills the commutator and $\mathrm{tr}(P_kF_{v_k}Q_k) = \mathrm{tr}(Q_kP_kF_{v_k})$. □

Corollary 3 (Kirchhoff junctions: off-loop derivative defect)**.** *If q is continuous at v and $\sum_e \partial_e q = 0$ over the incident edges, then for a loop arriving along a and departing along b*

$$F_v = \mathrm{i}\begin{pmatrix} 0 & \Phi_v \\ \bar{\Phi}_v & 0 \end{pmatrix}, \qquad \Phi_v := \sum_{e \notin \{a,b\}} \partial_e q\,(v), \tag{7}$$

the sum of outward derivatives on the off-loop edges. At degree two this is zero and a smooth ring is isospectral. At degree three it is the derivative on the remaining edge. The physical outward mass current is $j_e = 2\,\mathrm{Im}(\bar{q}\,\partial_e q)$, so its off-loop sum is $2\,\mathrm{Im}(\bar{q}\Phi_v)$; Φ_v itself is not the mass current. Nonzero F_v may change the trace, but is not a sufficient condition for doing so.

Proof. By continuity the diagonal entries of V_a^{in} and V_b^{out} agree. The off-diagonal difference is $\mathrm{i}(d_a^{\mathrm{in}} - d_b^{\mathrm{out}}) = -\mathrm{i}(\partial_a q + \partial_b q) = \mathrm{i}\sum_{e \notin \{a,b\}} \partial_e q$. □

Corollary 4 (Fixed gluing and local compatibility)**.** *A time-independent invertible crossing matrix C is locally compatible with the auxiliary time evolution precisely when $V_b^{\mathrm{out}}C = CV_a^{\mathrm{in}}$. Generic Kirchhoff boundary jets at a vertex of degree at least three do not satisfy this additional matrix relation. This is a statement about local Lax compatibility, not a proof that the total trace defect is nonzero for every solution or every fixed gluing.*

Proposition 5 (Nonzero vertex defect with conserved loop spectrum)**.** *Fix $c > 0$ and $0 \le s < c^2$. Let f_+ be the positive periodic solution of*

$$f_+'' = 2c^2 f_+ - 2f_+^3, \qquad f_+(0) = c, \quad f_+'(0) = s, \qquad L_s = \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sqrt{c^2 + s\sin\theta}}. \tag{8}$$

At $s = 0$ take the constant solution and $L_0 = \pi/c$. On a theta graph use two edges of length L_s, with $f_1(x) = f_+(x)$ and $f_2(x) = f_+(L_s - x)$, and a third edge of any length with

$f_3 = c$. *Then* $q_e(x,t) = \mathrm{e}^{2\mathrm{i}c^2 t} f_e(x)$ *solves NLS with Kirchhoff conditions at both vertices. Every travel-adapted loop spectrum is conserved, although the loop using edges* 1 *and* 3 *has nonzero vertex defects for* $s > 0$.

Proof. The first integral is $(f_+')^2 = s^2 - (f_+^2 - c^2)^2$, whose turning points $\sqrt{c^2 - s}$ and $\sqrt{c^2 + s}$ are positive. Substitution $f_+^2 = c^2 + s\sin\theta$ gives the period in (8). The outward derivatives at the two vertices are $(s, -s, 0)$ and $(-s, s, 0)$, respectively, so their sums vanish and all endpoint field values agree. Substitution of the standing-wave ansatz gives (1). With $G(t) = \mathrm{diag}(\mathrm{e}^{\mathrm{i}c^2 t}, \mathrm{e}^{-\mathrm{i}c^2 t})$ every spatial matrix satisfies $U_e(t) = G(t)U_e(0)G(t)^{-1}$, hence every edge and loop transport is conjugated by $G(t)$. For the loop using edges 1 and 3, the off-loop derivative is $-s\mathrm{e}^{2\mathrm{i}c^2 t}$ at the first vertex and $s\mathrm{e}^{2\mathrm{i}c^2 t}$ at the second. Equation (7) therefore gives nonzero defects for $s > 0$, whose trace pairings cancel by Theorem 2. All mass currents are zero since the profiles are real. □

This family separates vertex curvature from loss of isospectrality. It establishes no claim of standard integrability for arbitrary theta-graph data. It instead shows exactly why a curvature diagnostic must be paired with the global balance law, rather than treated as a binary integrability verdict.

Remark 6 (Mirror symmetry)**.** On a theta graph with edges e_1, e_2, e_3 between A and B, take the loops $\gamma_1 = e_1 e_2^{\leftarrow}$ and $\gamma_2 = e_3 e_2^{\leftarrow}$ and the commutator trace $\kappa = \mathrm{tr}(H_{\gamma_1} H_{\gamma_2} H_{\gamma_1}^{-1} H_{\gamma_2}^{-1}) - 2$, which vanishes if and only if the two holonomies share an eigenline [15, 16]. If e_1 and e_3 have equal length and the data are mirror symmetric, the fields on e_1 and e_3 remain equal for all time, $T_1 = T_3$, $H_{\gamma_1} = H_{\gamma_2}$ and $\kappa \equiv 0$. The individual traces can nevertheless drift; the plotted theta run illustrates trace drift and is not the mirror-symmetric case: $\kappa \equiv 0$ is a statement about symmetry, not about isospectrality.

4. DRESSED MONODROMY FROM VERTEX TIME TRANSPORTS

For each germ g at a vertex v define the time transport $W_g(t) \in \mathrm{SL}(2, \mathbb{C})$ by

$$\partial_t W_g = V_g(v,t)\, W_g, \qquad W_g(0) = \mathbf{1}, \tag{9}$$

with V_g the time-Lax matrix of the germ (V^{in} or V^{out} according to its direction of travel). The *vertex insertion* for a crossing that arrives along a and departs along b is

$$D_v[b \leftarrow a](t) := W_b(t)\, W_a(t)^{-1}, \qquad \partial_t D_v = V_b^{\mathrm{out}}\, D_v - D_v\, V_a^{\mathrm{in}}, \quad D_v(0) = \mathbf{1}. \tag{10}$$

D_v is the unique solution of the crossing equation with initial value $\mathbf{1}$. For a specified initial gluing C, the solution is instead $D_{v,C} = W_b C W_a^{-1}$. The endpoint quotient in (10) is not in general the laboratory-frame ordered exponential of $V_b^{\mathrm{out}} - V_a^{\mathrm{in}}$: the two noncommuting time evolutions must be retained. It compensates the crossing defect and generally depends on the field history. The *dressed loop transport* of the directed loop γ based at v_0 is

$$\mathcal{T}_\gamma(t) := D_{v_0}[e_1 \leftarrow e_n]\, T_n\, D_{v_{n-1}}[e_n \leftarrow e_{n-1}]\, T_{n-1} \cdots D_{v_1}[e_2 \leftarrow e_1]\, T_1. \tag{11}$$

Theorem 7 (Dressed monodromy)**.** *For any vertex conditions, $\mathcal{T}_\gamma(t) = W_{g_0}(t)\, \mathcal{T}_\gamma(0)\, W_{g_0}(t)^{-1}$, where g_0 is the departing germ of e_1 at v_0. Consequently $\operatorname{tr} \mathcal{T}_\gamma(\lambda, t)^n$ is conserved for every n and λ, and if two directed loops γ, γ' are based at the same vertex and depart along the same germ g_0, every joint conjugation invariant of the pair $(\mathcal{T}_\gamma, \mathcal{T}_{\gamma'})$ is conserved; in particular*

$$\hat{\kappa}(\lambda) := \operatorname{tr}\left(\mathcal{T}_\gamma \mathcal{T}_{\gamma'} \mathcal{T}_\gamma^{-1} \mathcal{T}_{\gamma'}^{-1}\right) - 2 \tag{12}$$

is a constant of the motion, and the two dressed transports share an eigenline at time t if and only if they do at time 0.

Proof. For every edge traversed in its direction of travel, $T_k(t) = W_{\mathrm{in},k}(t)\, T_k(0)\, W_{\mathrm{out},k-1}(t)^{-1}$, where $W_{\mathrm{in},k}$ is the time transport of the arriving germ of e_k at v_k and $W_{\mathrm{out},k-1}$ that of the departing germ of e_k at v_{k-1}: both sides solve (3) with the same initial value. Inserting this and (10) into (11), every interior factor $W_{\mathrm{out},k} W_{\mathrm{in},k}^{-1} \cdot W_{\mathrm{in},k} T_k(0) W_{\mathrm{out},k-1}^{-1}$ telescopes, leaving $W_{g_0}(t)\, T_n(0) \cdots T_1(0)\, W_{g_0}(t)^{-1}$, and $\mathcal{T}_\gamma(0) = T_n(0) \cdots T_1(0)$ since all $W(0) = \mathbf{1}$. The remaining statements are immediate, using that all matrices have unit determinant (V is traceless) so that the Fricke criterion [15] applies to the pair. □

Remark 8 (The base point is a germ)**.** Two loops leaving v_0 along different edges are conjugated by different matrices, W_g and $W_{g'}$, and their commutator trace need not be conserved (Fig. 4(b)); the change of base germ is itself a vertex insertion $D_{v_0}[g' \leftarrow g]$. This is the concrete form in which a vertex of degree three enters: the natural base point of the dressed holonomy is a germ, not a vertex.

Remark 9 (What is and is not decided)**.** $\hat{\kappa}$ tests simultaneous reducibility of the dressed pair: it vanishes precisely when the two $\mathrm{SL}(2,\mathbb{C})$ matrices share an eigenline. Its conservation follows from the common conjugation, irrespective of whether the NLS vertex condition is integrable in a standard sense. Because the dressing uses time history, this conservation alone does not establish an instantaneous local invariant, a commuting hierarchy of charges, or solvability by inverse scattering. Those are separate, unproved requirements. The construction retains the defect rather than requiring the original connection to be flat; the local compatibility tests below determine whether additional structure can close without history.

5. LOCAL INTEGRABLE JUNCTIONS: COMPATIBILITY CONDITIONS AND SCOPE

Definition 10 (Local vertex Lax data)**.** A *local transmission* from germ a to germ b is a generically invertible matrix $K(\lambda,t)=\sum_{k=0}^{n}\lambda^k K_k(t)$ whose coefficients depend on the instantaneous boundary fields and a specified finite set of their derivatives, and which satisfies

$$\partial_t K = V_b K - K V_a \qquad \text{identically in } \lambda. \tag{13}$$

This is the local Lax-compatibility criterion used here. It is a necessary ingredient in a defect-based integrable junction, rather than a definition that alone proves standard Hamiltonian or inverse-scattering integrability. Vertex degrees of freedom, if included, must be specified separately.

The exact solution of (13) is $K(t)=W_b(t)K(0)W_a(t)^{-1}$. Thus a local K realizes the history compensation with initial gluing $C=K(0)$. In general $K(t)K(0)^{-1}\neq W_bW_a^{-1}$; equality requires further commutation or normalization conditions. Replacing each insertion by $D_{v,C}$ preserves the conjugation proof of Theorem 7, now with the chosen C included in the initial loop product.

Proposition 11 (Degree-two positive control: the jump defect)**.** *Let u,v be the arriving and departing fields, with derivatives u_x,v_x in their directions of travel, and set $K=\lambda K_1+K_0$. The highest orders of* (13) *give*

$$K_1=\mathrm{diag}(a,b),\quad \partial_t a=\partial_t b=0, \qquad K_0=\begin{pmatrix} c_1(t) & \frac{\mathrm{i}}{2}(au-bv)\\ \frac{\mathrm{i}}{2}(b\bar{u}-a\bar{v}) & c_2(t)\end{pmatrix}. \tag{14}$$

The functions c_1, c_2 need not be constants. For $b \neq 0$, the upper off-diagonal equations imply

$$v_x = \frac{a}{b}u_x - \frac{2\mathrm{i}}{b}(c_1 u - c_2 v), \tag{15}$$

$$v_t = \frac{a}{b}u_t + G, \qquad G = \frac{2}{b}(c_1 u_x - c_2 v_x) - \frac{\mathrm{i}}{b}(|u|^2 + |v|^2)(au - bv). \tag{16}$$

The remaining equations, including evolution of c_1, c_2, are retained in the full condition $\partial_t K_0 = R_v K_0 - K_0 R_u$, where $R_q = \mathrm{i}\left(\begin{smallmatrix} |q|^2 & q_x \\ \bar{q}_x & -|q|^2 \end{smallmatrix}\right)$. A known integrable positive control is

$$K = \lambda \mathbf{1} + \frac{\mathrm{i}}{2}\begin{pmatrix} -\Omega & u - v \\ \bar{u} - \bar{v} & \Omega \end{pmatrix}, \qquad \Omega = \sqrt{\alpha^2 - |u - v|^2}, \quad \alpha > 0, \tag{17}$$

on a real branch with $|u - v| < \alpha$. It yields the Corrigan–Zambon conditions

$$u_x - v_x = \Omega(u + v), \qquad u_t - v_t = \mathrm{i}\Omega(u_x + v_x) + \mathrm{i}(u - v)(|u|^2 + |v|^2), \tag{18}$$

and $\det K = \lambda^2 + \alpha^2/4$ *[10].*

Proof. Write $V_q = -2\mathrm{i}\lambda^2\sigma_3 + 2\lambda Q_q + R_q$, with $\sigma_3 = \mathrm{diag}(1, -1)$ and $Q_q = \left(\begin{smallmatrix} 0 & q \\ -\bar{q} & 0 \end{smallmatrix}\right)$. The orders λ^3 and λ^2 force the diagonal leading coefficient and the displayed off-diagonal entries of K_0. The diagonal entries at order λ cancel, so a, b are time independent; the off-diagonal entries give (15). Order zero is the full matrix equation stated above, whose upper entry gives (16). For (17), set $a = b = 1$, $c_1 = -\mathrm{i}\Omega/2$ and $c_2 = \mathrm{i}\Omega/2$ to obtain (18). Differentiating the square-root constraint gives $\Omega_t = \mathrm{Im}((\bar{u} - \bar{v})(u_x + v_x))$, which supplies the diagonal equations. The lower entries are the conjugate counterparts and the determinant follows directly. These calculations retain the field-dependent diagonal terms required by the established defect. □

Theorem 12 (Three germs: necessary shared-germ compatibility)**.** *Suppose two degree-one local transmissions carry arriving germs* 1 *and* 3 *into the same departing germ* 2*, with nonzero b_1, b_3. Write their time conditions as $\partial_t q_2 = \rho_i \partial_t q_i + G_i$, where $\rho_i = a_i/b_i$ and G_i is (16) for that pair. Their simultaneous compatibility requires*

$$\rho_1 \partial_t q_1 - \rho_3 \partial_t q_3 + G_1 - G_3 = 0. \tag{19}$$

After substitution of NLS this is an equation involving second spatial derivatives of the incoming fields. It can be an additional restriction, or can follow from the other vertex equations and their time derivatives. With identical incoming germs and identical transmission data it is an identity. This test alone does not exclude local integrable junctions at degree three.

Proof. Equate the two expressions for $\partial_t q_2$ from Proposition 11. NLS gives $\partial_t q_i = \mathrm{i}(q_{i,xx} + 2|q_i|^2 q_i)$, yielding the stated second-derivative form. If both incoming fields and all their required jets agree, and both transmissions use the same coefficient functions, the terms cancel. For general data one must impose the spatial equations, the diagonal evolution equations and the differential consequences of all boundary conditions before testing independence of (19). Counting the displayed equations without this reduction would not prove a no-go result. □

Smooth Kirchhoff solutions themselves obey time-differentiated continuity, hence restrictions on endpoint second derivatives. The appearance of such derivatives is therefore not by itself an integrability obstruction. Conversely, a symmetric cancellation of (19) does not prove that either individual transmission exists. Both directions of this distinction matter.

Proposition 13 (Exact recurrence at arbitrary spectral degree)**.** *For $K = \sum_{k=0}^{n} \lambda^k K_k$, set $K_j = 0$ outside $0 \le j \le n$. Equation (13) is equivalent to, for $0 \le j \le n+2$,*

$$\partial_t K_j = -2\mathrm{i}[\sigma_3, K_{j-2}] + 2(Q_v K_{j-1} - K_{j-1} Q_u) + R_v K_j - K_j R_u. \tag{20}$$

For $n \ge 1$, $K_n = \mathrm{diag}(a, b)$ with $\partial_t a = \partial_t b = 0$, and $(K_{n-1})_{12} = \mathrm{i}(au - bv)/2$, $(K_{n-1})_{21} = \mathrm{i}(b\bar{u} - a\bar{v})/2$. Also $\det K$ is independent of time. If three polynomial transmissions obey the additional cocycle condition $K_{ab} K_{bc} K_{ca} = \mathbf{1}$, every determinant is a nonzero constant in λ, so every positive-degree leading coefficient is singular.

Proof. Collect the coefficients of λ^j in the polynomial matrix equation. The two top orders give the stated leading and next coefficients. In the diagonal part of order n, the term $2(Q_v K_{n-1} - K_{n-1} Q_u)$ cancels $R_v K_n - K_n R_u$, giving $\partial_t K_n = 0$. For generic invertible K, Jacobi's formula gives $\partial_t \det K = \det K \ \mathrm{tr}(K^{-1} \partial_t K) = \det K \ \mathrm{tr}(V_b - V_a) = 0$; polynomial continuation extends this identity to all λ. Taking determinants of the cocycle condition gives a product of three nonzero polynomials equal to one, so each is constant. A nonsingular leading coefficient of positive degree would give a nonconstant determinant, a contradiction. □

A singular leading coefficient is not a proof of decoupling: a matrix can have singular leading coefficient and invertible constant determinant, for example $\left(\begin{smallmatrix} 1 & \lambda \\ 0 & 1 \end{smallmatrix}\right)$. This example is not claimed to solve the NLS transmission equation. At degree one, the rank-one case $K_1 = \mathrm{diag}(a, 0)$ under the cocycle assumption has $(K_0)_{22} = 0$ and $\det K = -a^2 u\bar{v}/4$; its derivative relations still involve $(K_0)_{11}$, which may depend on both fields. They therefore do

not establish independent Robin half-lines without additional restrictions. At degree zero a diagonal constant gluing imposes fields and derivatives related by a common phase (for nonzero fields with the uniform NLS nonlinearity), a restricted proportional-profile sector. The recurrence is a constructive test for higher degree, while a complete classification remains open.

Remark 14 (Joint vertex data and comparison with existing integrable sectors)**.** An alternative ansatz uses a matrix $\mathbf{K}(\lambda)$ on $\bigoplus_e \mathbb{C}^2$ satisfying a reflection-type equation $\partial_t \mathbf{K} = \mathbf{V}(\lambda)\mathbf{K} - \mathbf{K}\mathbf{V}(-\lambda)$, with $\mathbf{V} = \bigoplus_e V_e$ [12]. This is a specified ansatz, not the most general vertex object. Equation counts do not determine the rank of its compatibility conditions. Coupling solutions, restricted-data solutions, and extensions with vertex degrees of freedom as in type-II defects [11] require separate analysis.

Remark 15 (Caudrelier and the transparent rescaling control)**.** Caudrelier's star-graph inverse-scattering construction uses outward coordinates and matrix boundary conditions [4]. Its linearizable boundary conditions include disconnected Robin problems and coupling constructions on specified symmetric data subspaces. These provide comparison sectors; they do not constitute a classification of every nonlinear local junction in Definition 10.

The transparent networks of Sobirov *et al.* use nonlinearities β_e, profiles $q_e = \sqrt{\beta_0/\beta_e}\, f$ and a sum rule at the vertex [9]. For the spatial matrix $U_e = \begin{pmatrix} -\mathrm{i}\lambda & q_e \\ -(\beta_e/2)\bar{q}_e & \mathrm{i}\lambda \end{pmatrix}$ one has $U_e = D_e U_f D_e^{-1}$, with $D_e = \mathrm{diag}(d_e, d_e^{-1})$ and $d_e^2 = \sqrt{\beta_0/\beta_e}$. On equal oriented intervals carrying the same profile this gives $T_e = D_e T_f D_e^{-1}$. Products using inverse transports and the induced crossing matrices $D_{e'} D_e^{-1}$ then cancel to the identity, whereas arbitrary identity gluing need not do so. Figure 5 checks this change-of-basis identity. It uses ordinary inverse-path transport, rather than the travel-adapted reverse-edge convention of Theorems 2 and 7; its identity product is not a claim that all travel-adapted loops are trivial. The gluing and orientation convention are data that must be specified together.

6. NUMERICAL VERIFICATION

The finite-difference runs impose second-order Kirchhoff conditions (Appendix A), with $h = 0.02$, $dt = 2 \times 10^{-4}$; finer runs use $h = 0.01$, $dt = 5 \times 10^{-5}$. Strang splitting combines an exact nonlinear phase with a Crank–Nicolson linear step. A bright soliton of amplitude

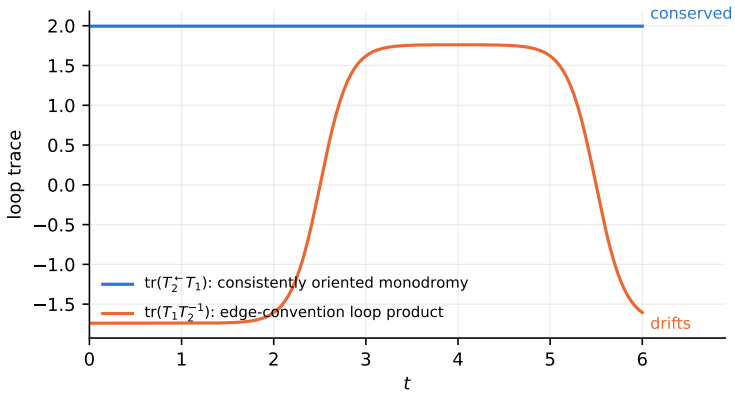


FIG. 2. Ring with two degree-two Kirchhoff vertices, soliton datum, $\lambda = 0.6$. The monodromy built from transfers in the direction of travel is conserved (drift 5×10^{-6}); the product of one transfer with the group inverse of the other is not (drift 3.5), because the inverse is the transfer of the reflected field at $-\lambda$ (Lemma 1).

1.6, velocity 2, on the length-10 edge reaches B at $t \approx 2.5$. Transfers use closed-form cell exponentials at $\lambda = 0.6$; vertex derivatives use one-sided four-point differences.

a. Independent algebra and standing-wave controls. Direct substitution of (17) into (13) on 30 seeded boundary-jet samples and four real or complex spectral values gives maximum matrix-entry residual 6.3×10^{-16}. For Proposition 5, with $c = 1$, $\lambda = 0.6$, third-edge length 1.1 and $s = 0, 0.15, 0.3, 0.6$, independent spatial integration at times 0 and 0.4 gives defect norms $0, 0.212, 0.424, 0.849$ while the loop-trace change stays below 2×10^{-15}. At $s = 0.6$ the two individual defect pairings have magnitude 0.0843 while their sum is below 9×10^{-12}. Thus the nonzero-curvature example tests cancellation rather than a diagnostic insensitive to all defects.

b. Ring (Fig. 2). Two edges of lengths 6 and 10 join A and B, both of degree two, so the graph is a smooth periodic ring of length 16. The consistently oriented monodromy $\mathrm{tr}(T_2^{\leftarrow} T_1)$ is conserved to 5×10^{-6} over the run, while the edge-convention product $\mathrm{tr}(T_1 T_2^{-1})$ swings from -1.74 to $+1.76$, a drift of 3.5. This is Lemma 1 made visible: the group inverse mixes the spectral problems at $\pm\lambda$.

c. Defect law on the theta graph (Fig. 3). Edges $(6, 10, 8)$ join degree-three vertices A, B. For $\gamma_1 = e_1 e_2^{\leftarrow}$, fourth-order time differences of trace samples 0.002 apart agree with (6) to a median 0.1% of the peak rate during vertex activity (0.14% at $h = 0.01$). Maximum

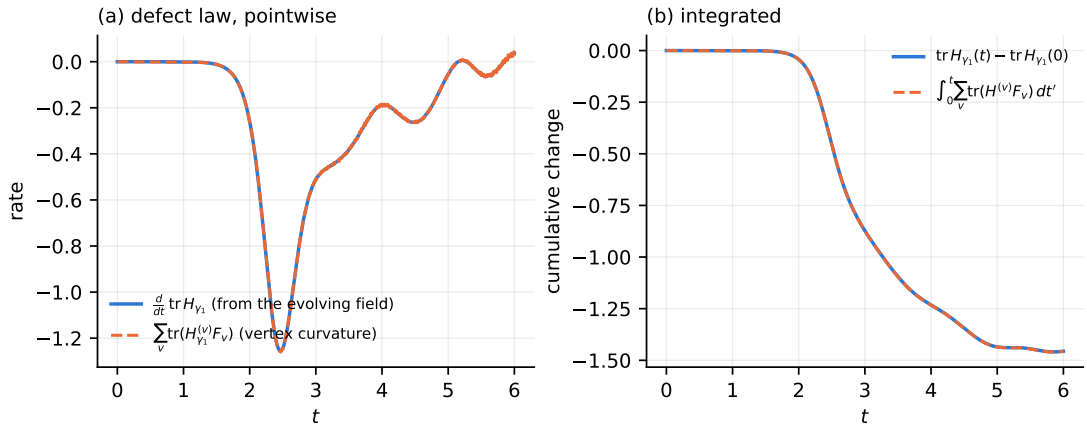


FIG. 3. Theta graph $(6, 10, 8)$, Kirchhoff vertices, soliton launched on e_2, $\lambda = 0.6$. (a) Rate of change of the loop trace computed from the evolving field versus the vertex-curvature sum $\sum_v \operatorname{tr}(H_{\gamma_1}^{(v)} F_v)$. (b) Cumulative change versus the integrated vertex-curvature sum. The trace starts to move when the soliton reaches the degree-three vertex and the field begins to leak into the third edge.

discrepancies are 1.1% at $h = 0.02$ and 1.8% at $h = 0.01$, mainly during late short-wavelength radiation. Integrating the curvature sum reproduces the total change -1.456 to 5×10^{-4} on both grids. Curvature-sum excursions reflect the discrete vertex error: the trace departs from a smooth curve by less than 2×10^{-5}, while the independently measured Kirchhoff sum has median residual 0.2–0.4% of the largest germ derivative (Appendix A). Corollary 3, $(F_B)_{12} = \mathrm{i}\partial_3 q$ and $(F_B)_{21} = \mathrm{i}\overline{\partial_3 q}$, holds to a median 0.5% ($h = 0.02$) or 0.7% ($h = 0.01$). Pointwise refinement errors are not monotone.

d. Dressed monodromy (Fig. 4). The germ time transports (9) are integrated at every time step from the vertex values of the field. The dressed trace of the loop $e_1 e_2^{\leftarrow}$ plotted in Fig. 4(a) is conserved to about 4.3×10^{-5}, against an undressed drift of 1.46. The dressed commutator trace $\hat{\kappa}$ of the two loops that depart A along the same edge e_2 is conserved to 2×10^{-6}; that of the two loops departing along e_1 and e_3 drifts by 1.8×10^{-2} (Remark 8); the undressed κ drifts by 1.0.

e. Rescaling control (Fig. 5). Three equal oriented intervals of length 1.3 use $\beta_e = (3, 4, 12)$, $\beta_0 = 2$, and $q_e = \sqrt{\beta_0/\beta_e}\, f$ for two specified profiles. The inverse-transport products $T_1 T_2^{-1}$ and $T_2 T_3^{-1}$ with identity gluing have commutator traces reaching approximately 5×10^{-4} over the plotted real spectral range. Inserting the induced crossing matrices reduces both products to identity, with commutator residuals at roundoff. This is a static auxiliary-basis

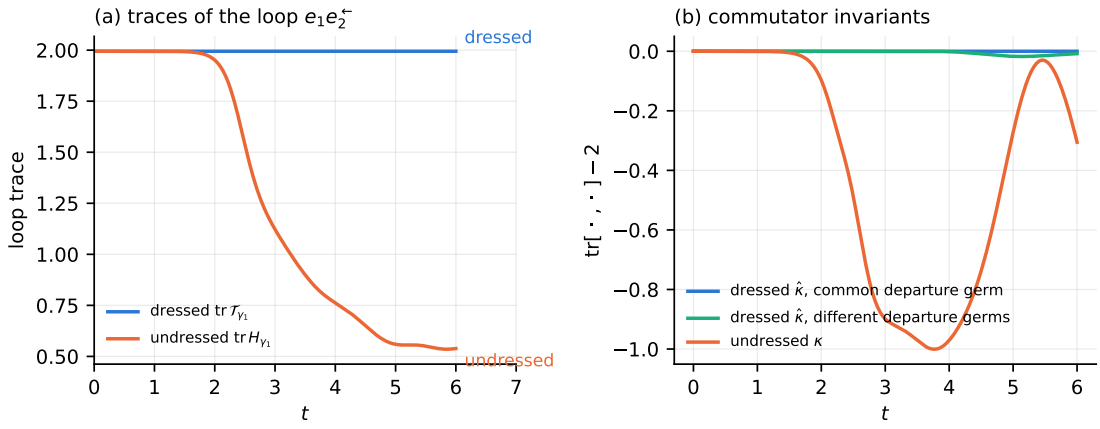


FIG. 4. Same run as Fig. 3. (a) Trace of the loop $e_1 e_2^{\leftarrow}$ with the vertex insertions (10) (dressed) and without (undressed). (b) Commutator traces: dressed, for two loops departing A along the same germ (conserved); dressed, for two loops departing along different germs (drifts, Remark 8); undressed κ, which is generated in real time as the soliton leaks into the third edge and is not an invariant.

control, not a simulation of a travel-adapted loop spectrum.

f. Soliton routing. The theta soliton splits at the degree-three vertex. Its off-loop derivative enters the spectral balance; the undressed commutator trace grows to order one while the common-germ dressed trace stays fixed within numerical error. The standing-wave family instead has nonzero derivative defects and zero spectral drift. Mirror symmetry gives equal loop transports and $\kappa = 0$, without requiring constant individual traces (Remark 6).

7. INTERPRETATION: VERTEX CURVATURE AND THE LOWEST COHERENCE OPERATION

On an edge, vanishing compatibility curvature of (U, V) is exactly NLS. At a vertex crossing, endpoint time matrices can disagree by F_v (Theorem 2); the theory permits this defect. Travel-adapted reversal is not inverse-path transport, so the matrix identities establish neither a fundamental-group representation nor a global surface-transport functor.

A resolved higher-gauge crossing can encode the mismatch in a 2-form B_v concentrated at the vertex world-line. For a differential crossed module $\mathfrak{h} \xrightarrow{t} \mathfrak{g}$, t maps degree-two data to ordinary curvature. *Fake flatness* means $F_A = t(B)$, with F_A the connection curvature [17, 18];

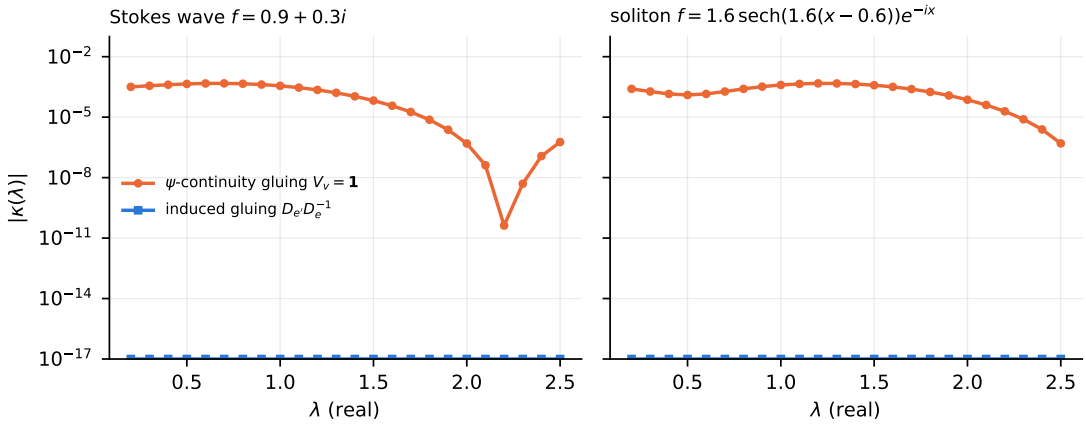


FIG. 5. Non-central rescaling control on equal oriented intervals, motivated by the transparent-field construction of Sobirov *et al.*. Inverse-transport products under identity auxiliary gluing give nonzero commutator traces (orange); induced gluing $D_{e'}D_e^{-1}$ gives identity products (blue, roundoff floor). These products use matrix inverses, not travel-adapted reversed transfers. Their distinction from Figs. 2–4 is deliberate.

the vertex coefficient obeys $t(B_v) = F_v$, with arriving-minus-departing sign. This controls nonzero curvature rather than imposing $F_v = 0$.

In a two-term L_∞ description, the lowest operation is the unary map $\ell_1 = t$: $F_v = \ell_1(B_v)$ is the lowest-rung relation. The binary operation encodes brackets and actions; higher homotopies require their own coherence equations. This retains the coherence-defect interpretation of [19]. The quotient $W_b W_a^{-1}$ supplies exact compensation and covariance, but a global surface holonomy needs a specified crossed module, compatible path/surface data and composition checks.

The inner crossed module $t = \mathrm{id}$ formally permits $B = F_A$; this alone proves no new integrability. A stronger categorical claim needs an instantaneous defect-sensitive invariant, a specified rung and interpolating parameter, a known integrable positive control, a separating example, and a prediction beyond history dressing. Here we supply the lowest relation, the degree-two positive control and the standing-wave separation. A full higher tower and local non-vacuous higher-rung invariant remain open. The matrix theorems do not depend on their construction.

8. CONCLUSION

The travel-adapted NLS loop trace obeys an exact balance law pairing loop transport with endpoint time-Lax defects. For Kirchhoff coupling, off-loop derivatives determine those defects. Soliton routing shows order-one spectral drift; the standing-wave family shows nonzero curvature with fixed spectra.

History insertions give simultaneous conjugation and conserved joint invariants for loops sharing a departure germ, without establishing standard integrability. Local polynomial transmissions face additional compatibility tests. The Corrigan–Zambon defect supplies a positive control; the shared-germ equation and arbitrary-degree recurrence constrain more complicated junctions, without a universal degree-three no-go theorem.

The higher-gauge reading retains non-flatness through $F_v = \ell_1(B_v)$. Local closure, composition and higher coherence with a defect-sensitive invariant remain tasks for a categorical construction. A graph inverse-scattering or Hamiltonian theory must reproduce the balance law and pass these further tests.

Appendix A: Numerical methods

Edges use uniform grids ($h = 0.02$ unless stated), with a shared endpoint degree of freedom q_v enforcing continuity. Interior nodes use the three-point Laplacian. Second-order one-sided derivatives impose the Kirchhoff (or δ) constraint $\sum_e (3q_v - 4q_e^{(1)} + q_e^{(2)})/(2h_e) = g\, q_v$ as a Crank–Nicolson row. Strang splitting combines this constrained linear step with the exact nonlinear phase. The constraint has $O(h^3)$ residual on exact linear star solutions; vertex accuracy is second order, but exact discrete unitarity is lost. Run mass drifts are 10^{-5} ($h = 0.02$) and 2×10^{-6} ($h = 0.01$), versus 10^{-11} for the replaced first-order lumped-mass scheme.

Controls on both schemes include a stationary soliton (single-edge mass drift 5×10^{-12}), a moving soliton (peak position exact to the grid, envelope error 6×10^{-3}), and the Holmer–Marzuola–Zworski δ-vertex transmission trend [8]. A soliton of amplitude 2 and velocity 4 reflects 0.0770 of its mass at a degree-three Kirchhoff star, agreeing to four digits between schemes at $h = 0.04, 0.02$. The linear limit is $1/9$; a linear wave packet agrees on both schemes. Independent four-point derivatives measure the degree-three Kirchhoff residual at

a median 0.2–0.4% of the largest germ derivative, versus 2% for lumped mass, setting the pointwise error floor.

Cell-midpoint transfers use $\exp(hU) = \cos(\omega h)\mathbf{1} + \omega^{-1}\sin(\omega h)U$, $\omega^2 = \lambda^2 + |q|^2$. Germ time transports use the closed-form traceless-matrix exponential at each midpoint time-Lax matrix; germ derivatives use one-sided four-point differences. Two-grid pointwise defect discrepancies are not monotone: these are finite-resolution checks of proved identities, not certified continuum bounds. Scripts and outputs reproducing the numbers and figures are openly available in the accompanying Zenodo dataset [20].

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